

Zihan Liu

Male | Han Chinese | +86 139-7530-4333 | Email: JamesL@stu.xjtu.edu.cn | GitHub: github.com/topjames666 | Portfolio: topjames666.github.io/personal-web



Education

Xi'an Jiaotong University| C9 League | School of Energy and Power Engineering | Master's (Recommended for Admission)
2025.09 – Present

- Research Focus: Vertical AI Agent

East China University of Science and Technology| 211 | Process Equipment and Control Engineering | Bachelor's
2021.09 – 2025.06

Projects (More at topjames666.github.io/personal-web)

DeepAgents Multi-Agent Web Scrapping System| Independent Developer 2025.12 – Present

- Overview: A multi-agent system built with **DeepAgents** framework using **Orchestrator Pattern**, with 1 orchestrator agent coordinating 4 specialized sub-agents for fully automated web scrapping pipeline.
- Key Work: Built multi-agent workflow with dynamic decision-making and **fault-tolerant retry mechanism**, improving task completion rate from 60% to **95%+**; developed 8 **LangChain Tools** using **@tool decorator**; designed **Docker sandbox** for secure LLM-generated code execution; reduced end-to-end time from 10 min to **3 min**. Successfully scraped data from Douban Movies, Zhihu, and other websites.

Turbomachinery Intelligent Design Agent| Independent Developer (Master's Research) 2025.09 – Present

- Overview: A vertical AI agent built with **LangChain + LangGraph**, integrating **RAG knowledge enhancement** and professional design tools for turbomachinery engineers.
- Key Work: Designed local **RAG** solution (**Qdrant + Ollama**) with **5000+ pages** of technical documents, achieving **>80%** Q&A accuracy; integrated **3 MCP Servers** via langchain-mcp-adapters; built workflow with **>90%** autonomous planning rate; implemented **3-minute one-click deployment** with Docker Compose. System deployed in lab environment.

ReviewGo AI Literature Review System (Startup)| Project Lead & Full-Stack Developer 2025.12 – Present

- Overview: An **AI SaaS product** for researchers to generate academic literature reviews from PDF papers, reducing 2-4 weeks of work to hours.
- Key Work: Led full-stack development of **8 frontend pages (React 19 + shadcn/ui + Zustand)** with PDF upload, progress monitoring, and user center features; designed **multi-step LLM workflow** for PDF parsing and review generation; implemented automatic citation retrieval; built a 3-person team for FastAPI backend development.

Rehabilitation Danger Detection System| Project Lead 2024.01 – 2024.06

- Overview: A real-time danger posture recognition system using **YOLOv5** for rehabilitation patient safety monitoring.
- Key Work: Collected and annotated **3000+ images** of rehabilitation danger postures; trained YOLOv5 (**PyTorch**) for **500+ epochs**, achieving **F1-score 90%+** and **90% recognition accuracy**; approved by Haobo Medical for pilot testing.

Paper Finder Multi-Source Search System| Independent Developer 2026.01

- Overview: A frontend-backend separated academic paper search system supporting **CNKI + arXiv** with LLM-powered keyword expansion.
- Key Work: Implemented task creation with **Flask RESTful API + vanilla JavaScript**; unified multi-source search (**Playwright** for CNKI, official API for arXiv) with concurrent search achieving **3-5x speedup**; **≥90%** extraction success rate in **≤60 seconds** for 5 keywords.

Internship

Guangdong Telecom Planning & Design Institute (Changsha)| Smart City Project Intern 2023.07 – 2023.09

- Participated in smart city projects; conducted policy research, data analysis, and government reporting
- Collected smart city case studies and participated in proposal presentations for Kaifu District
- Managed data collection and location mapping for ferry crossings in Hubei Province

Honors & Awards

- **National Scholarship 2024**
- ECUST Outstanding Scholarship 2023, 2025
- "Internet+" Innovation Competition (Shanghai) **Silver**
- "Challenge Cup" Entrepreneurship Competition **Shanghai Silver**
- "Shangtu Cup" Engineering Drawing Competition **Team 1st Prize, Individual 2nd Prize**
- Utility Model Patent (1st Inventor)
- Design Patent (2nd Inventor)